

AI + Python Prompt Ideas for Business Analytics Students

Data Exploration, Segmentation, and Modeling

Prepared for classroom use

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Why these prompts work

- They move students beyond coding into **business thinking**.
- They encourage an **end-to-end workflow**: explore, segment, model, explain, recommend.
- They can be adapted for retail, banking, telecom, e-commerce, HR, and operations datasets.
- They support both **technical skills** and **manager-facing communication**.

Data exploration prompt ideas

1. Customer behavior exploration

Using Python, explore a retail customer dataset to identify patterns in spending, purchase frequency, product preferences, and seasonality. Summarize the main business insights and suggest three manager actions.

2. Sales performance diagnosis

Analyze a sales dataset and determine which variables most influence revenue differences across regions, products, and teams. Use visualizations and descriptive statistics to explain where performance is strongest and weakest.

More data exploration ideas

3. Churn signal discovery

Perform exploratory data analysis on a churn dataset. Detect missing data issues, correlations, unusual patterns, and early warning signs of customer loss. Explain which variables appear most useful for prediction.

4. Operational efficiency review

Use Python to examine an operations dataset and find bottlenecks, inefficiencies, and variability in process times. Present findings in a form that would help managers improve productivity.

Segmentation prompt ideas

5. Customer segmentation for marketing

Use Python to segment customers based on purchasing behavior, demographics, and engagement. Compare at least two clustering methods and recommend the most useful segmentation for a marketing strategy.

6. Product segmentation

Segment products by sales volume, profitability, return rates, and customer ratings. Identify which product groups should be promoted, redesigned, or discontinued.

7. Store or branch segmentation

Cluster stores or branches using operational and financial KPIs. Explain how the groups differ and how the segmentation could improve resource allocation.

Additional segmentation idea

8. Employee performance segmentation

Use Python to segment employees based on productivity, attendance, training, and performance indicators. Discuss how segmentation could support HR decisions while also noting ethical concerns.

Teaching note: This prompt is useful for discussing both analytics value and responsible use of AI in business settings.

Predictive modeling prompt ideas

9. Churn prediction model

Build a Python model to predict customer churn. Compare at least three algorithms, evaluate their performance, and explain which features contribute most to the predictions.

10. Sales forecasting

Develop a Python model to forecast future sales using historical business data. Compare a baseline forecasting approach with a more advanced model, and discuss which is more practical for managers.

More predictive modeling ideas

11. Credit risk modeling

Create a predictive model in Python to classify customers by credit risk. Evaluate accuracy, precision, recall, and interpretability, then explain which model is most suitable in practice.

12. Lead conversion prediction

Build a model to predict which leads are most likely to convert into customers. Use the results to design a prioritization strategy for the sales team.

Advanced applied prompt ideas

13. From exploration to action

Use Python to explore the dataset, identify business problems, segment observations if useful, and then build a predictive model. Present the analysis as if you were advising an executive team.

14. Manager-friendly AI analytics

Perform an end-to-end project in Python: data cleaning, exploratory analysis, segmentation, predictive modeling, and recommendations. Keep the final explanation understandable for non-technical managers.

More advanced applications

15. Explainable business AI

Build a predictive business model in Python, with a special focus on explainability. Use feature importance or similar methods to explain the model's decisions to stakeholders.

16. Scenario-based analytics

Analyze the dataset and simulate business scenarios such as reducing churn, increasing marketing spend, or improving delivery times. Recommend the highest-impact intervention.

Classroom framing ideas

- “You are a junior business analyst working for a retail chain.”
- “A telecom company has hired you to reduce churn.”
- “A bank wants a credit risk model that managers can understand.”
- “An e-commerce manager wants customer segments for a new campaign.”

Why use roles? Realistic framing helps students write stronger prompts, justify methods, and communicate recommendations more clearly.

Top five prompts to start with

- ① Customer behavior exploration
- ② Customer segmentation for marketing
- ③ Churn prediction model
- ④ Sales forecasting
- ⑤ From exploration to action

These five give students a balanced introduction to **EDA, clustering, prediction, forecasting, and executive communication**.

Suggested assignment extension

- Ask students to choose one prompt and refine it for a specific industry.
- Require both:
 - a Python notebook or script
 - a short business presentation of results
- Grade on: data quality checks, analytical reasoning, model evaluation, and business recommendations.

Thank you

These prompts can be used for
in-class labs, homework, mini-projects, or capstone preparation.